Implement a basic classification technique using Scikit-learn for categorical outcome prediction.

Theory

• **Library Import:** Scikit-learn is used to implement classification algorithms. The `KNeighborsClassifier` class can be imported from `sklearn.neighbors` to perform classification.

• **Dataset Preparation:** Input features (independent variables) and target labels (categorical outcomes) are prepared using NumPy arrays or Pandas DataFrames.

• **Data Splitting:** The `train_test_split()` function from `sklearn.model_selection` is used to divide the dataset into training and testing sets.

• **Model Creation:** A classification model is created using the `KNeighborsClassifier()` function, which classifies new data based on the nearest data points.

• **Model Training:** The `fit()` method is used to train the classifier using the training dataset.

• **Prediction:** The `predict()` method is used to predict categorical outcomes for new input data.

• **Model Evaluation:** The `accuracy_score()` function from `sklearn.metrics` is used to calculate the accuracy of the classifier by comparing predicted labels with actual labels.

• **Classification Output:** The model predicts discrete categories or class labels, such as Yes/No, Pass/Fail, or different product categories.

Conclusion

Thus, a basic classification technique was implemented using Scikit-learn to predict categorical outcomes. The model was trained using labeled data, used to predict class labels, and evaluated using classification accuracy.